PONTIFICIA UNIVERSIDAD CATÓLICA DEL PERÚ

1INF03 - Análisis de datos
2026-2

SEMANA 2:

STEP 3
"PRE-PROCESAMIENTO: LIMPIEZA DE DATOS"

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stefany.neciosup@pucp.edu.pe

9 STEPS

Fuente: Fayyad, Piatetsky-Shapiro, & Smyth (1996)

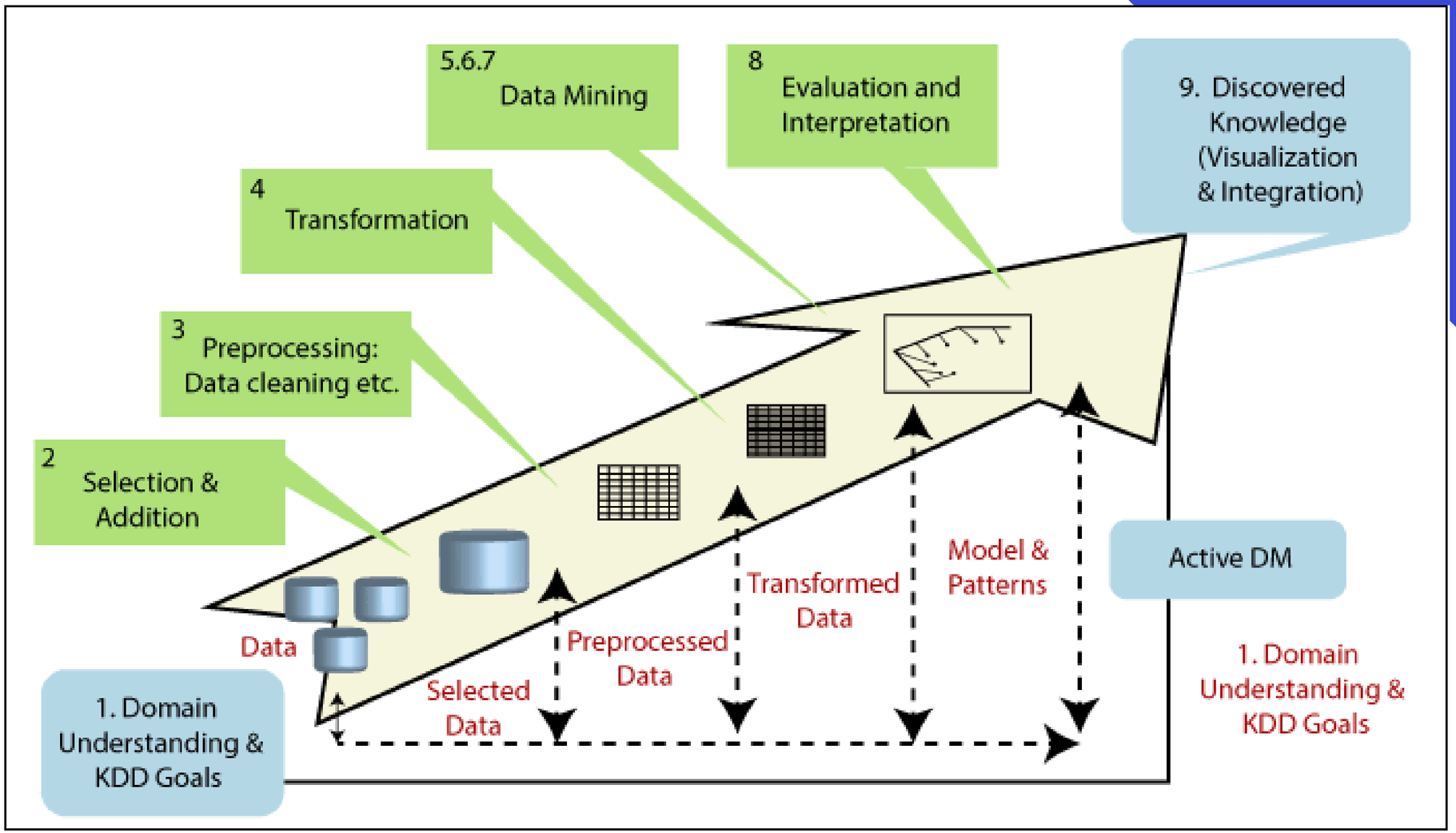


Imagen: https://www.javatpoint.com/kdd-process-in-data-mining

9 STEPS

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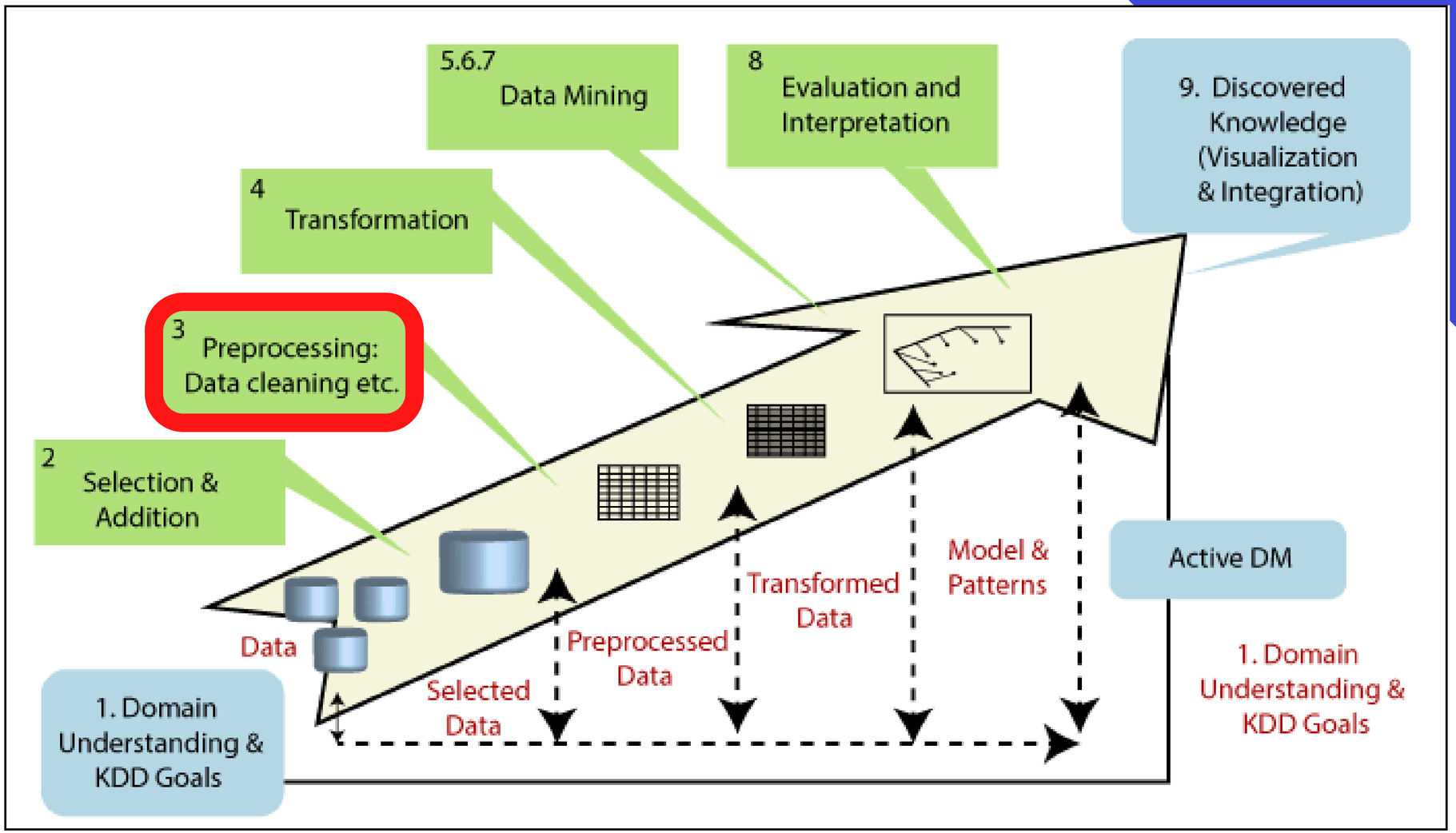
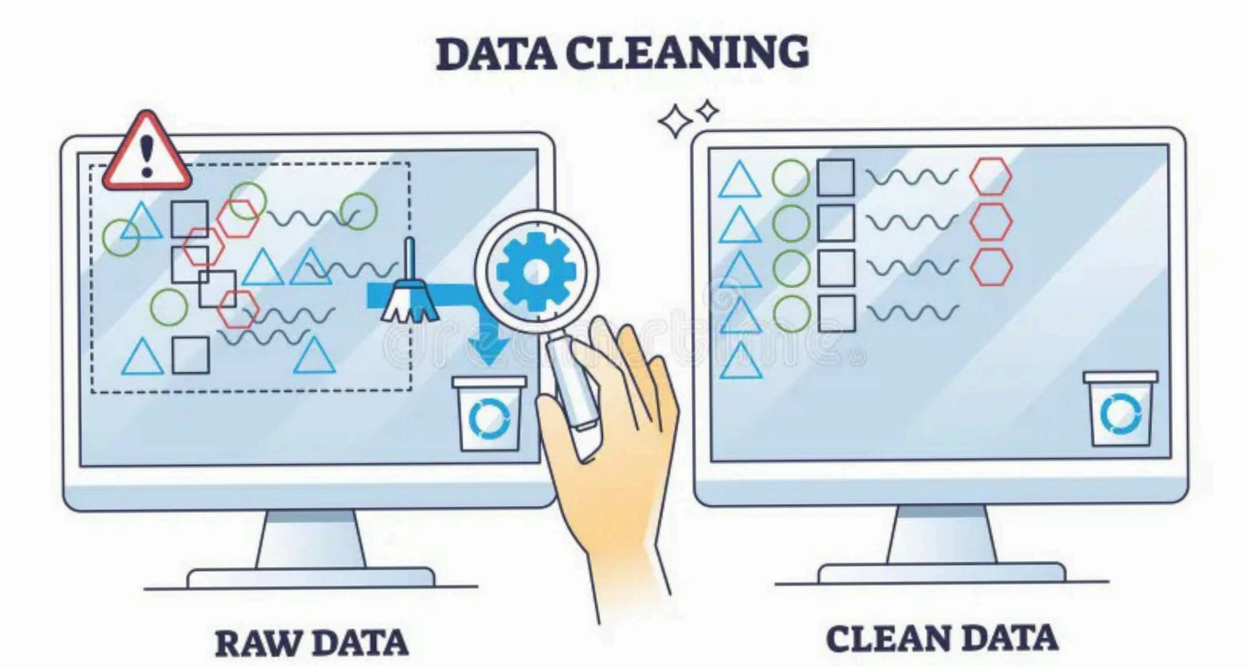


Imagen: https://www.javatpoint.com/kdd-process-in-data-mining

DATA CLEANING: ¿POR QUÉ ES IMPORTANTE?



DATOS DUPLICADOS

- Duplicados completos

  Son filas completamente iguales en todas las columnas.

  `df.duplicated()`

- Duplicados parciales

  Son filas solo duplicadas en una o más columnas clave, como ID y fecha

  `df.duplicated(subset=['ID', 'fecha'])`

DATOS VACÍOS

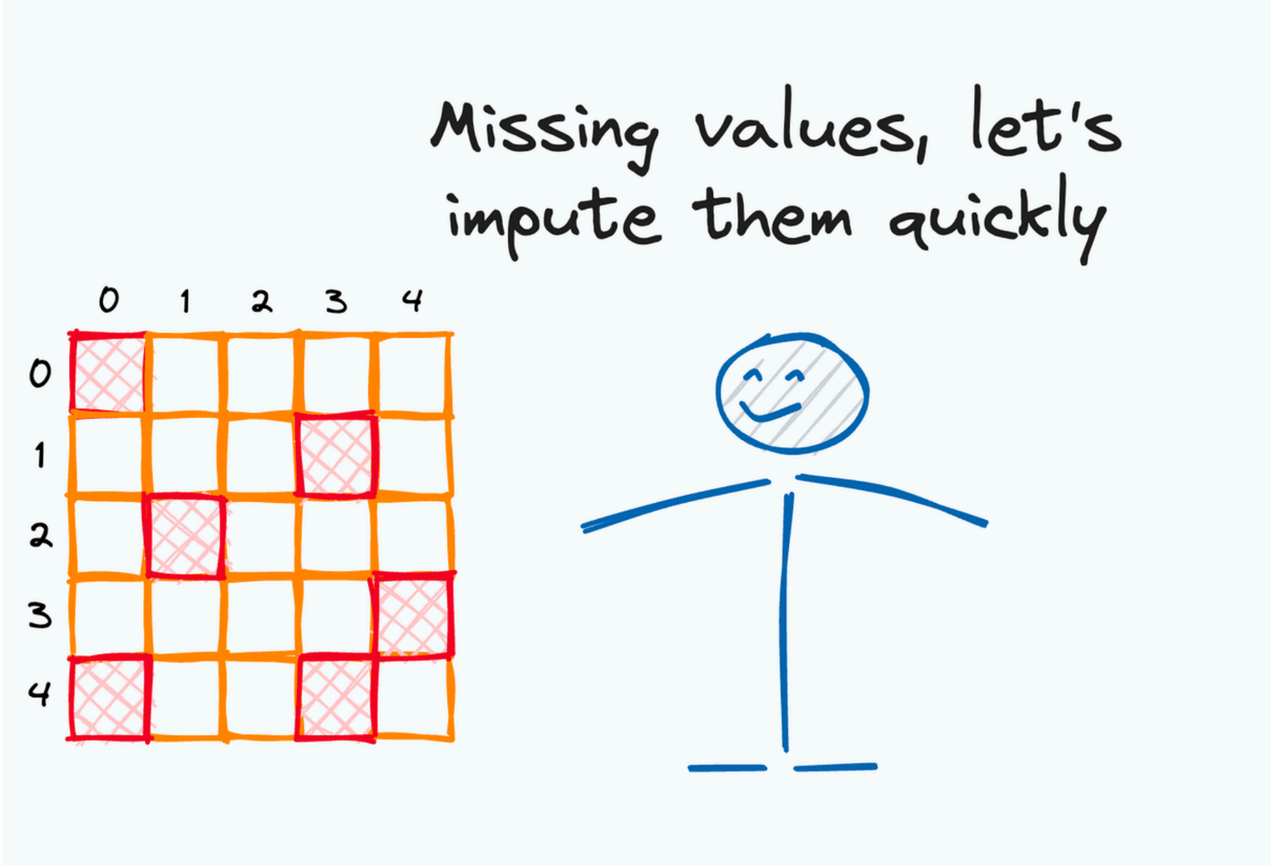
Datos vacíos: Empty data

Celdas vacías visualmente, pueden contener un espacio en blanco, una cadena vacía "", o pueden no tener nada. Deben convertirse a NA o NaN.

" " : espacio en blanco

"" : cadena vacía

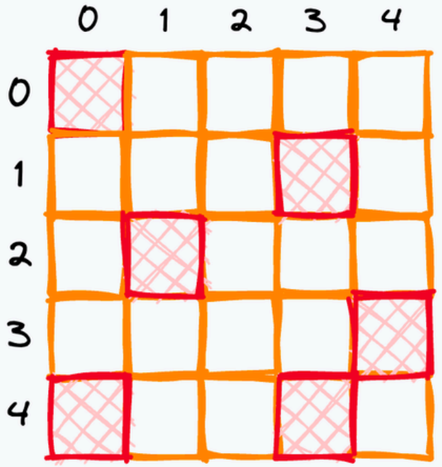
NA / NaN : sin dato



https://blog.dailydoseofds.com/p/3-types-of-missing-values



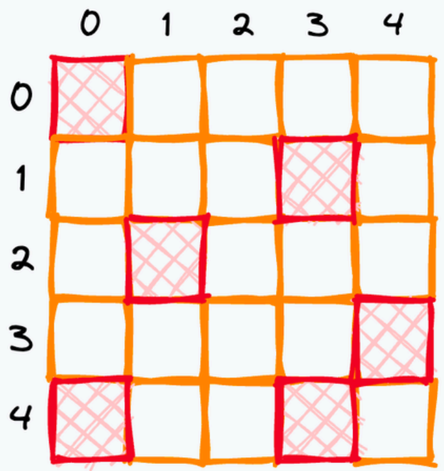
Missing values, let's impute them quickly







I must understand WHY do I have missing values before imputing them





https://blog.dailydoseofds.com/p/3-types-of-missing-values

Datos faltantes: Missing data

- MCAR -> Missing Completely at Random

  El dato falta completamente al azar, sin relación con otras variables ni con el valor perdido.

- MAR -> Missing at Random

  El dato faltante depende de otras variables observadas, pero no del valor que falta.

- MNAR -> Missing Not at Random

  El hecho de que el dato falte sí depende del valor que falta (no observable). Muy problemático.

Missing completely at random (MCAR)

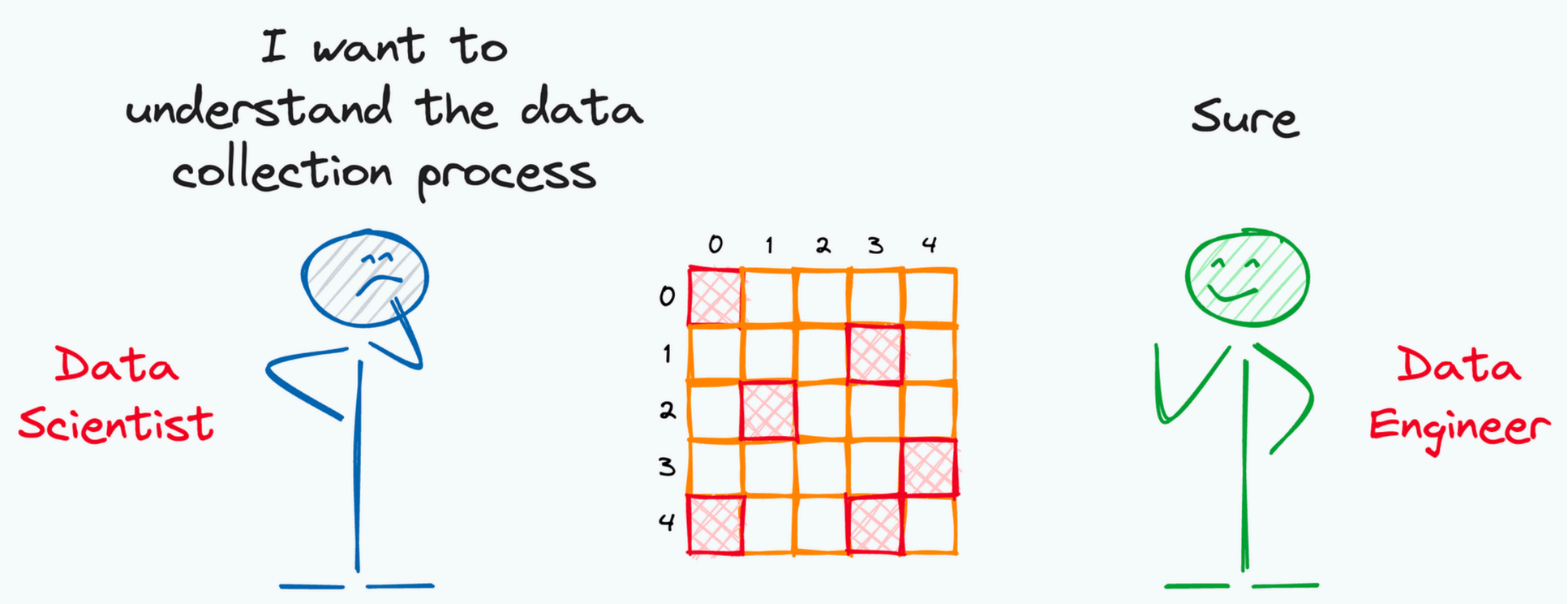
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https://blog.dailydoseofds.com/p/3-types-of-missing-values

Missing completely at random (MCAR)

Causa ----> Soluciones

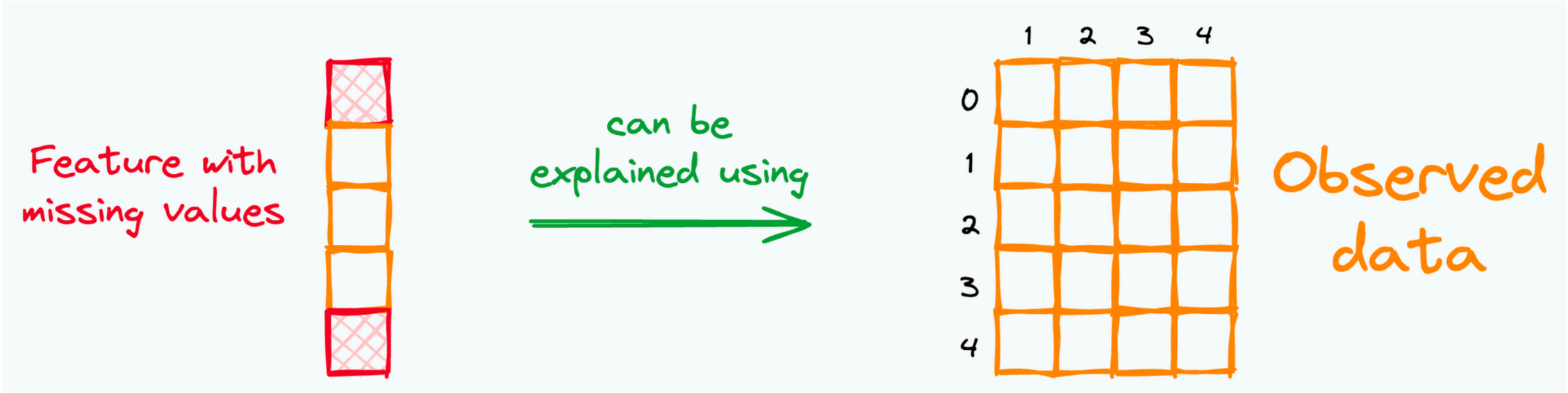
MCAR

- Eliminar filas
- Eliminar columnas
- Imputación simple o univariada

https://blog.dailydoseofds.com/p/3-types-of-missing-values

Missing at random (MAR)

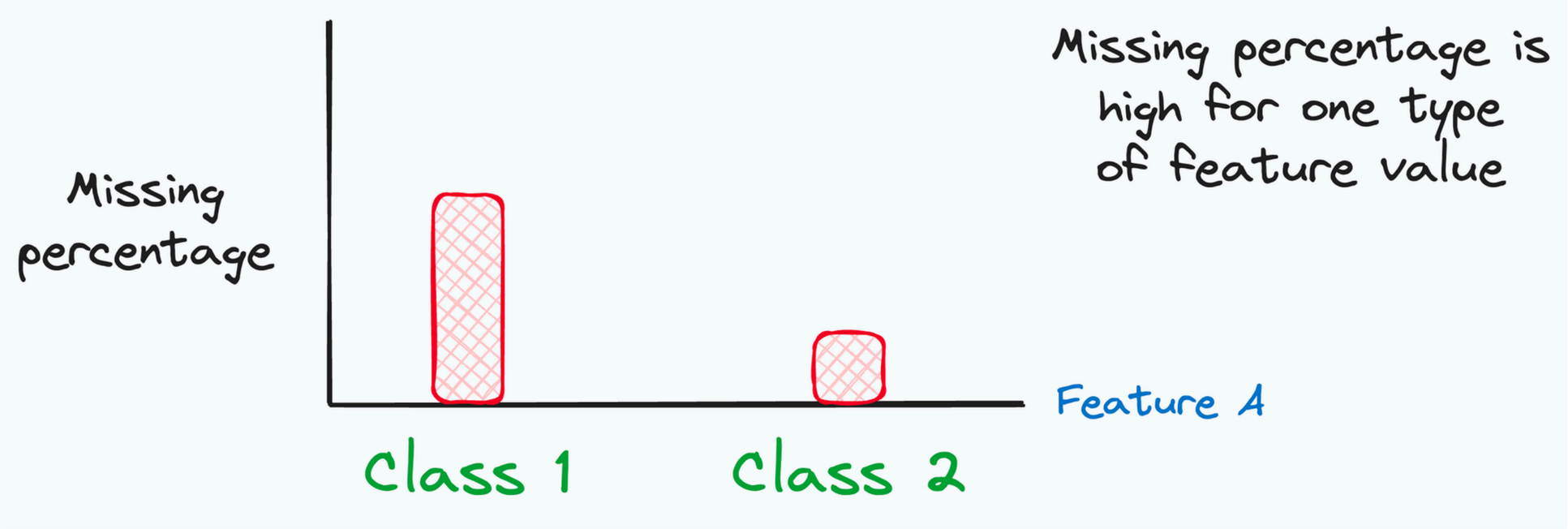
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https://blog.dailydoseofds.com/p/3-types-of-missing-values

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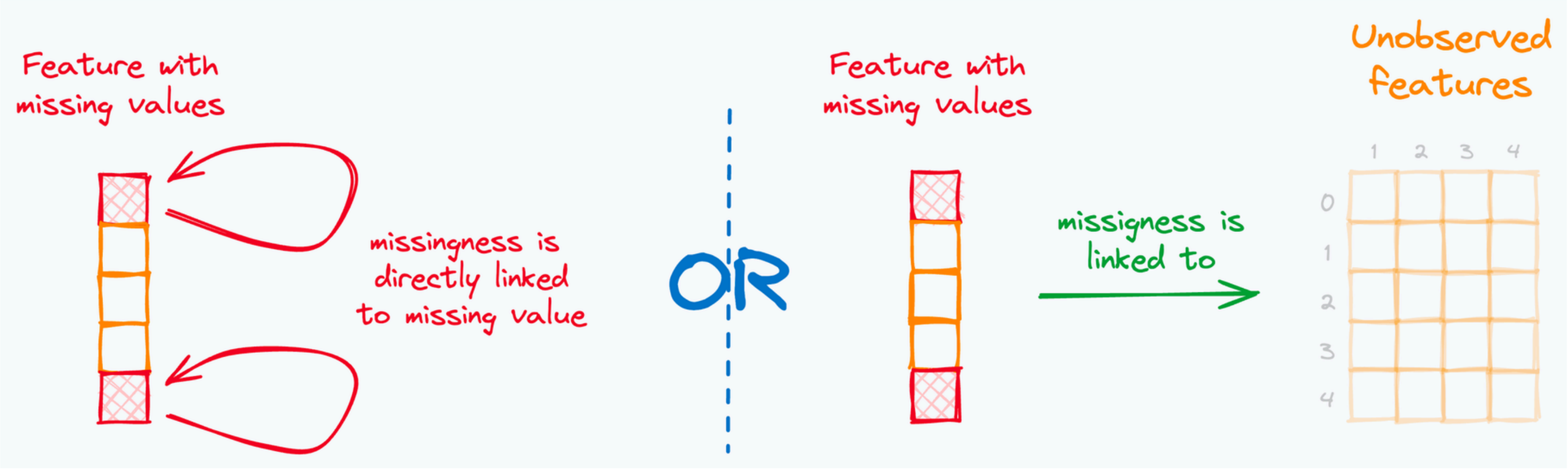
Causa ----> Soluciones

MAR

- Imputación condicional
- Imputación multivariada
- Modelos bayesianos o EM

Missing not at random (MNAR)

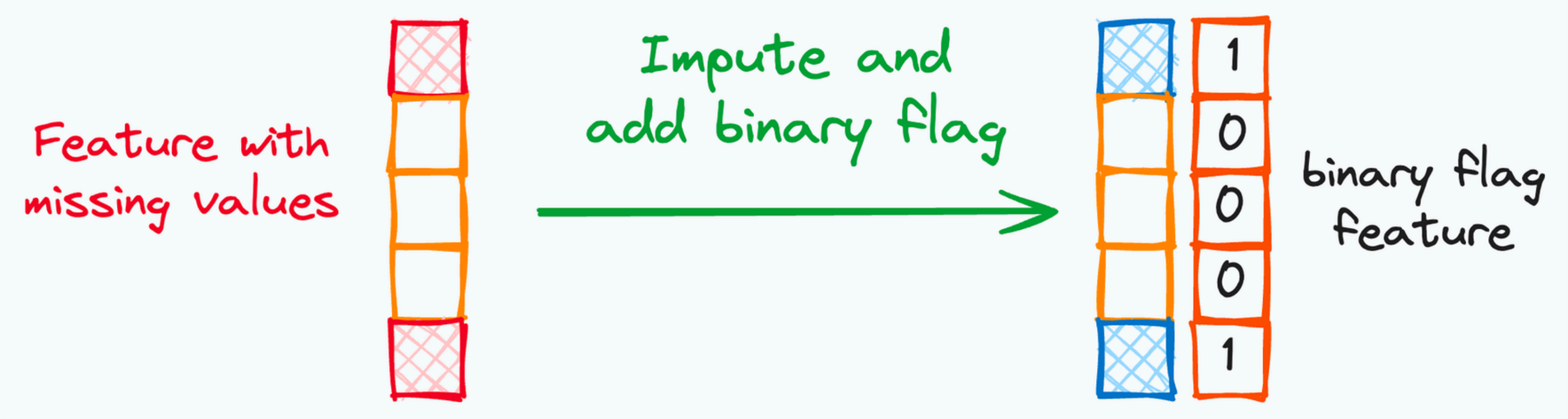
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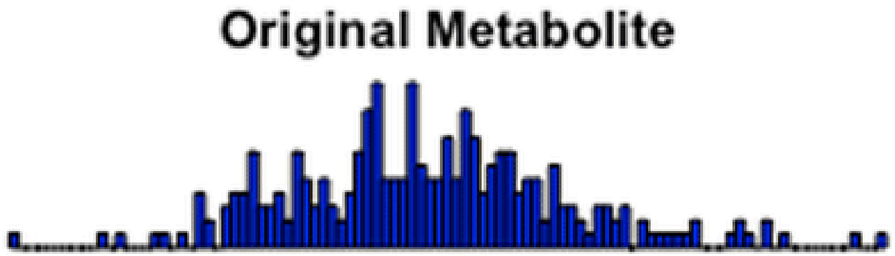
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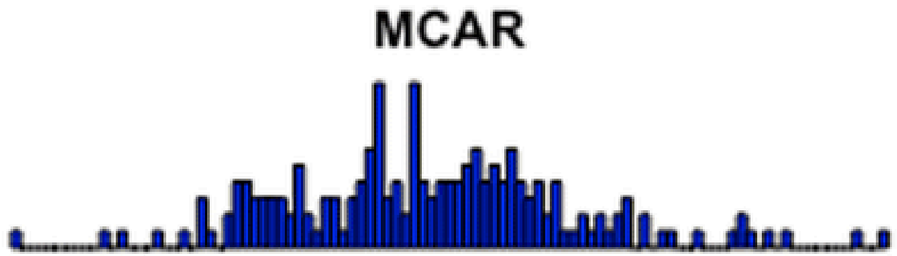
Causa ----> Soluciones

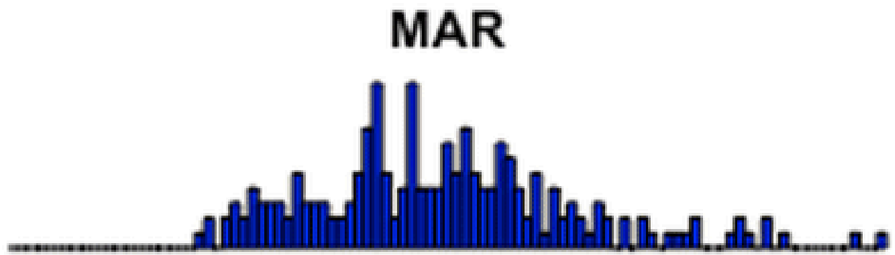
MNAR

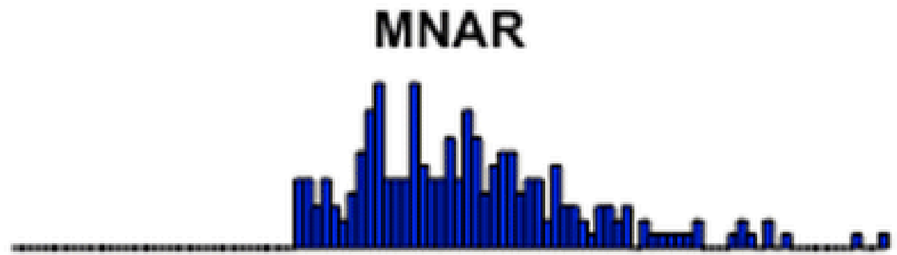
- Recuperar valores de fuentes externas
- Heckman imputation
- Mejorar mecanismos de recolección de datos

Simulation of missing values



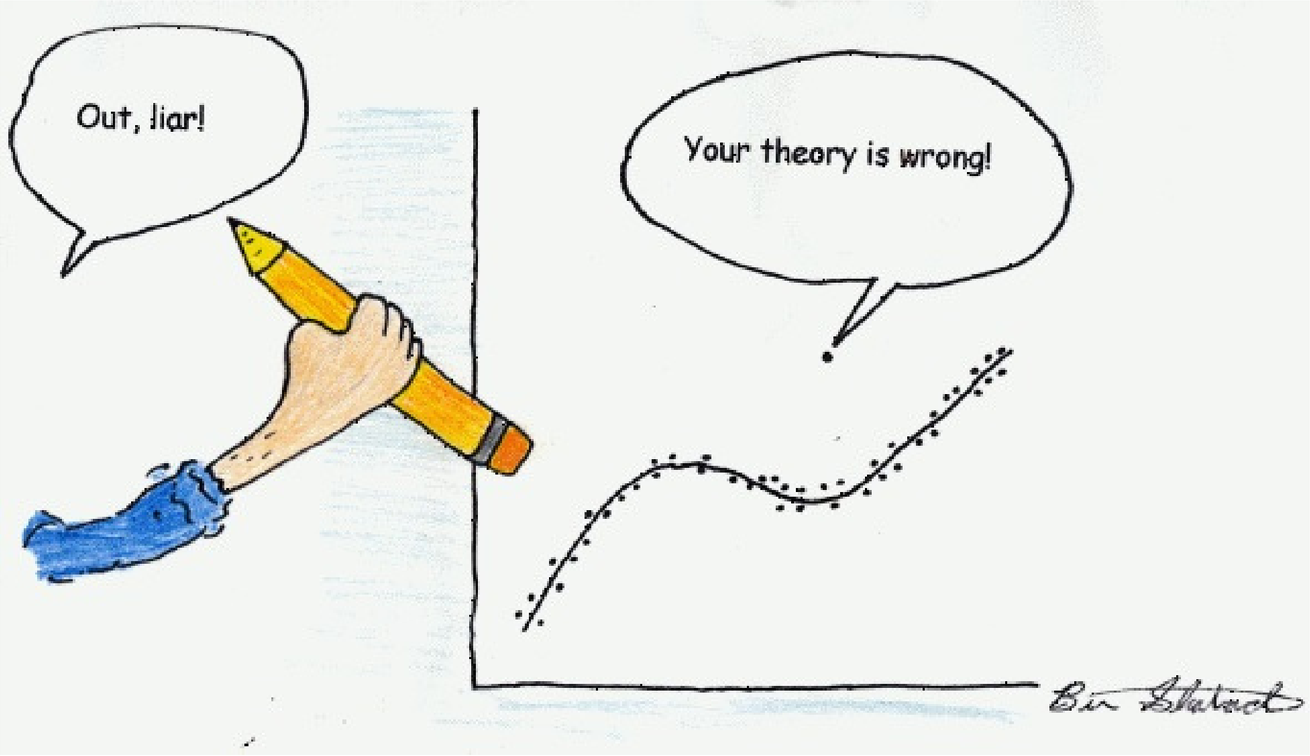




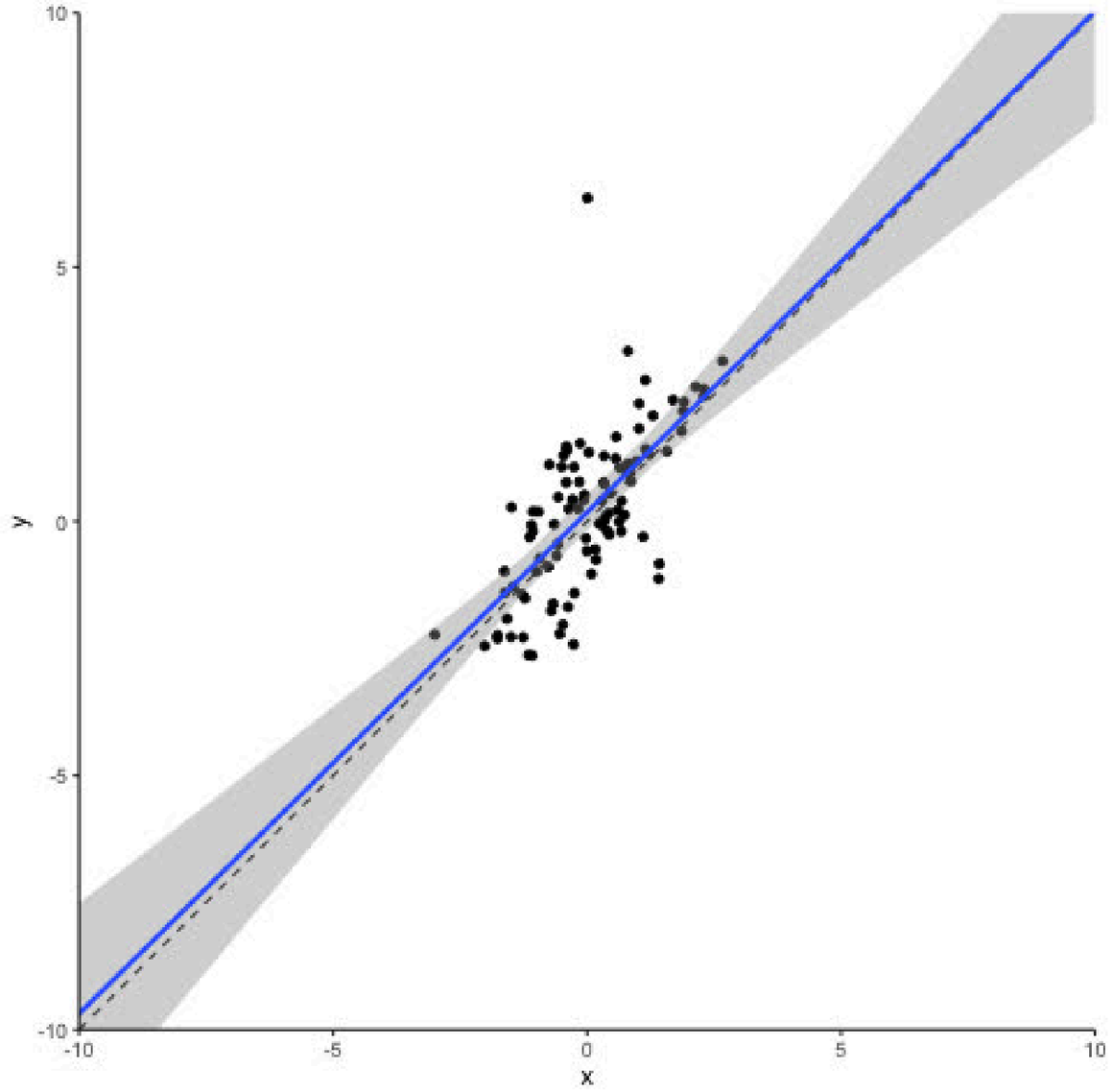


https://www.researchgate.net/figure/Distribution-based-examples-of-the-simulated-missing-mechanisms-Simulation-of-missing_fig5_336471983

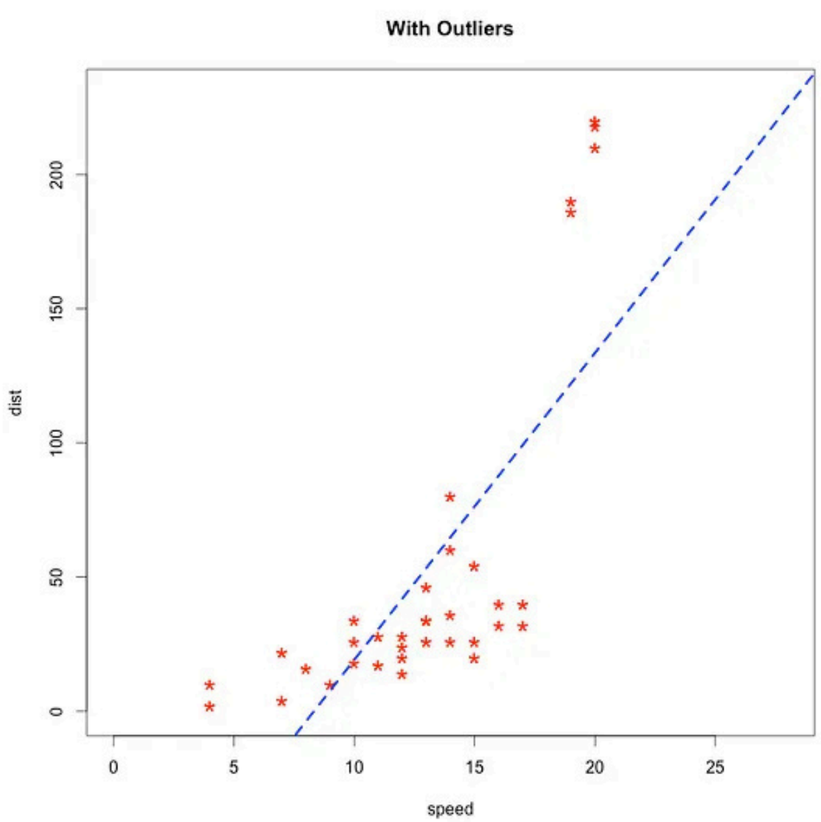
DATOS ATÍPICOS

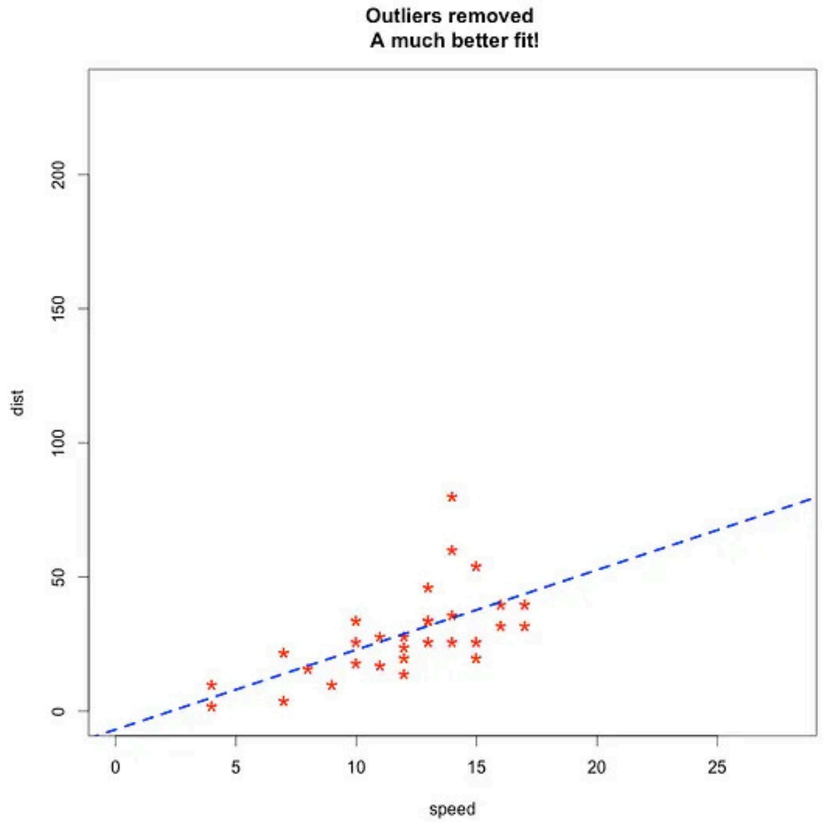


Simulation of outlier value



Simulation of outlier value





Opciones de tratamiento para datos atípicos

- Trimming

Eliminar todas las filas con valores atípicos.

- Capping

Reemplazar atípicos por valores límites por variable

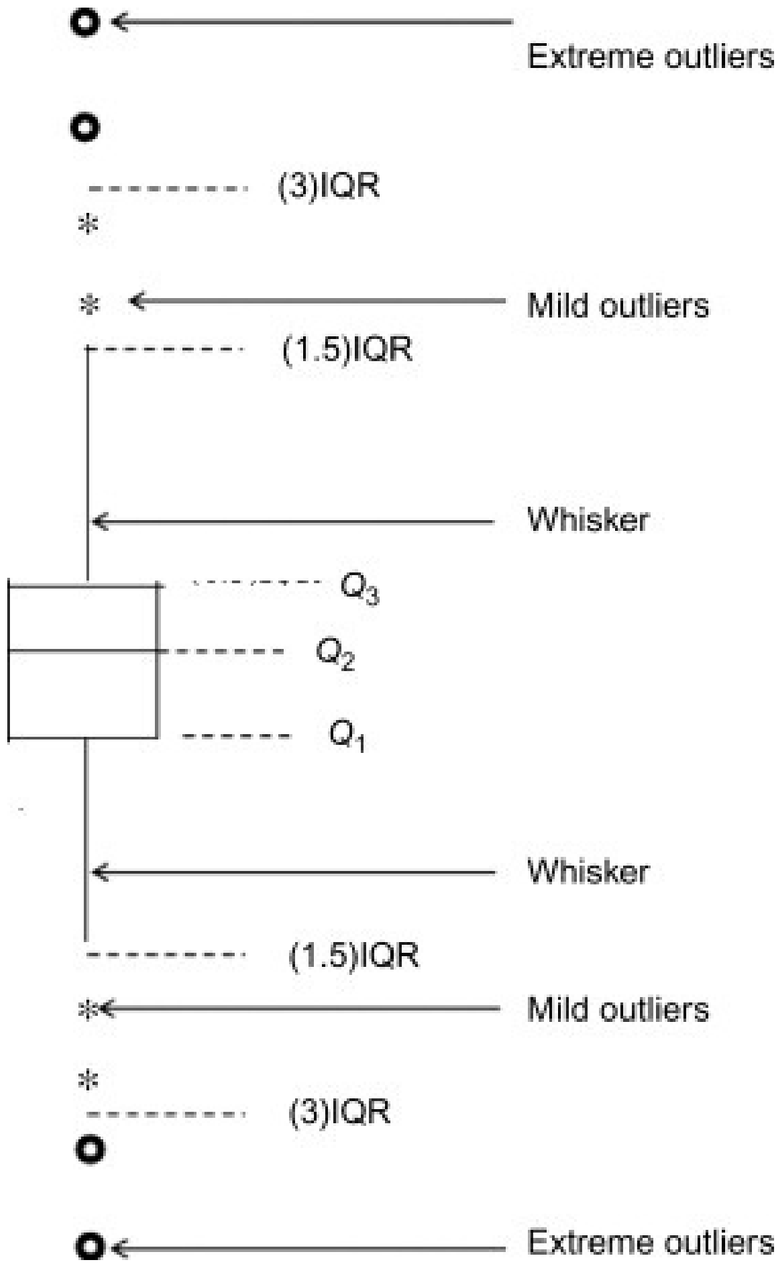
- Imputation

Reemplazar atípicos con técnicas de mputación.

- Transformation

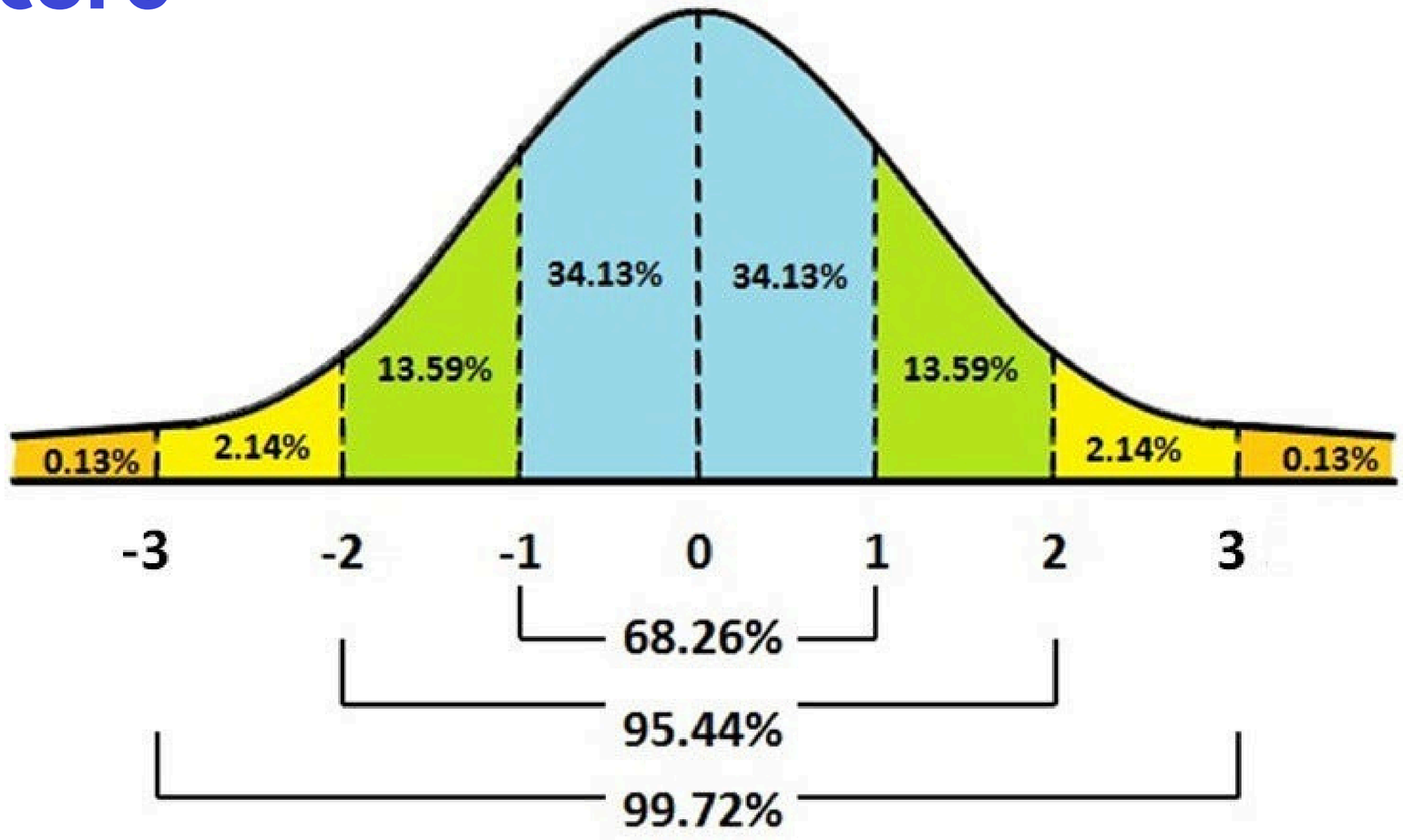
Aplicar transformaciones como log o raíz cuadrada

Capping
with IQR



https://www.sciencedirect.com/topics/mathematics/extreme-outlier

Capping
with Z-score



¿ Normalizar = Standarizar ?

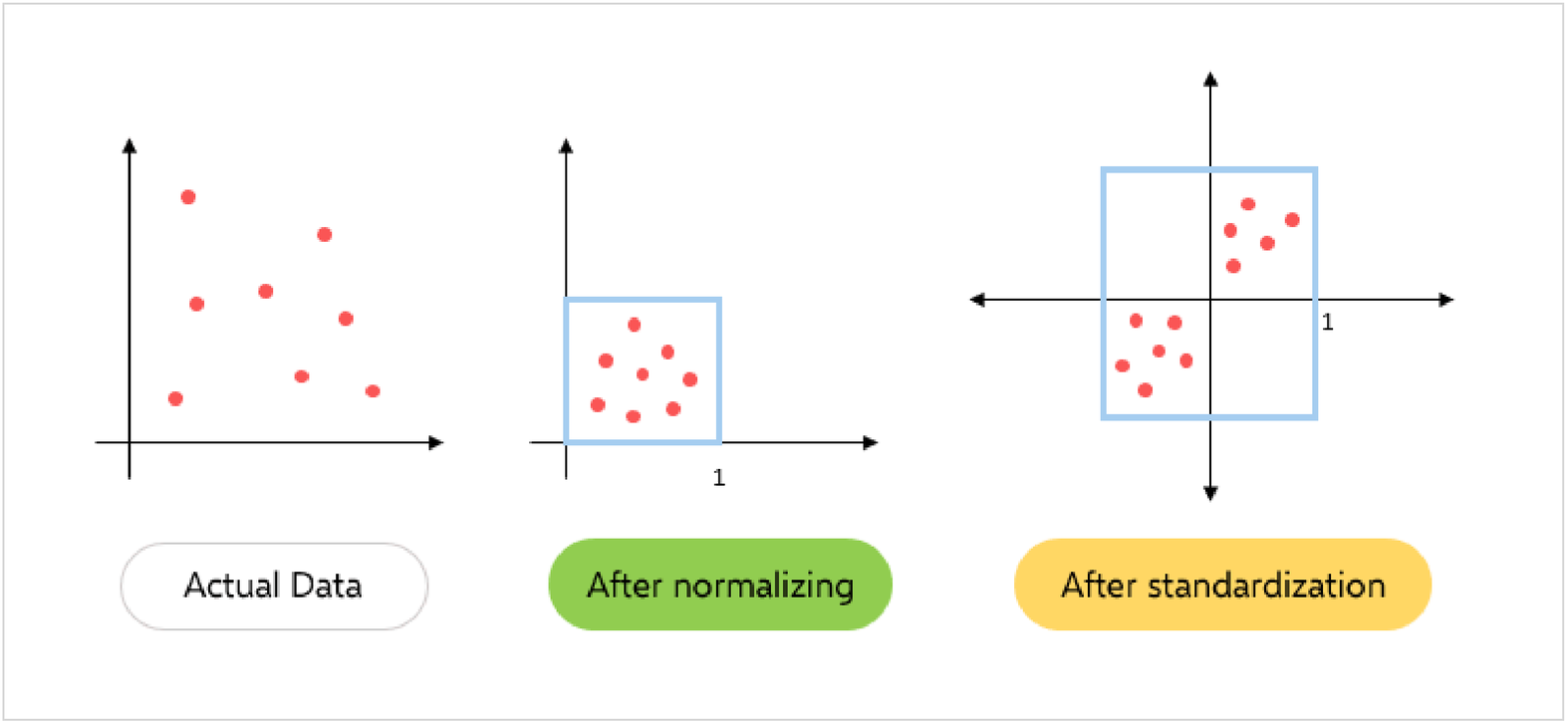
¿ Normalizar = Standarizar ?

Max-Min Normalization

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Standardisation (Z-score Normalization)

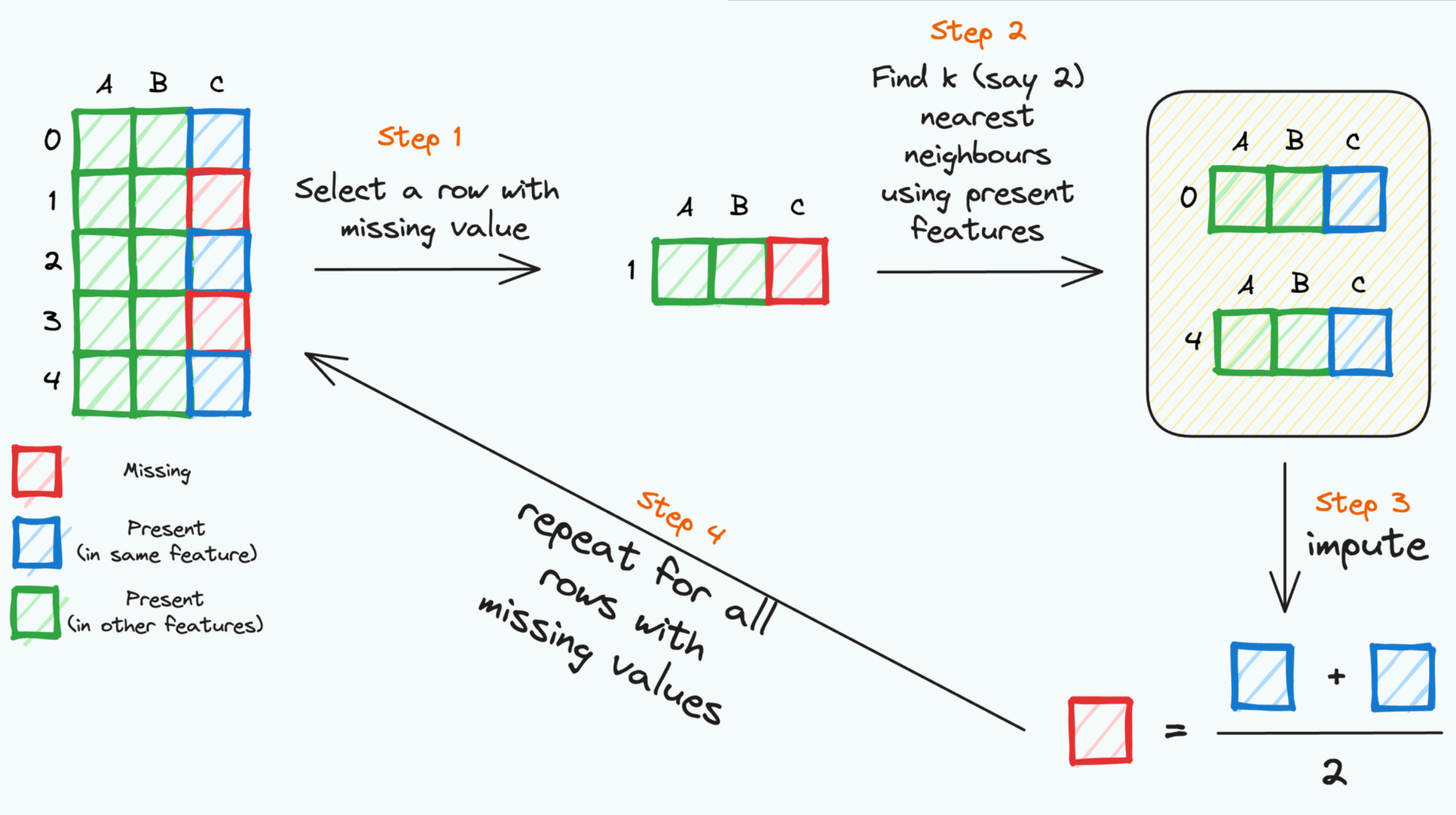
$$x_{\text{stand}} = \frac{x - \text{mean}(x)}{\text{standard deviation } (x)}$$



https://www.someka.net/wp-content/uploads/2020/08/difference-between-normalization-standardization-1.png

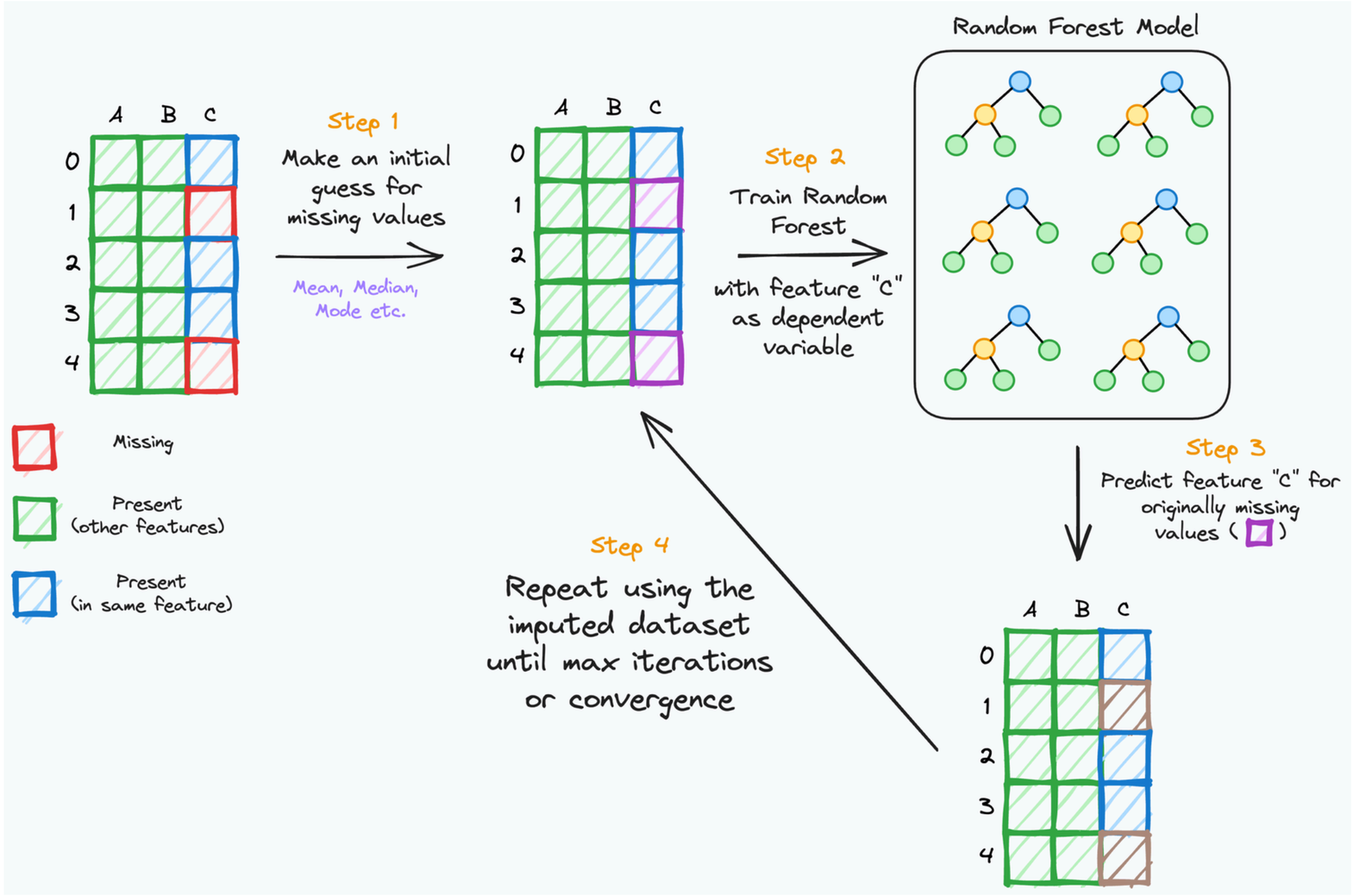
Imputation

with KNN Imputation



https://blog.dailydoseofds.com/p/3-types-of-missing-values

Imputation
with Miss Forest



https://blog.dailydoseofds.com/p/3-types-of-missing-values

How to handle outliers

errors or anomalies and
no significant size change



in target variable and want to
maintain data distribution



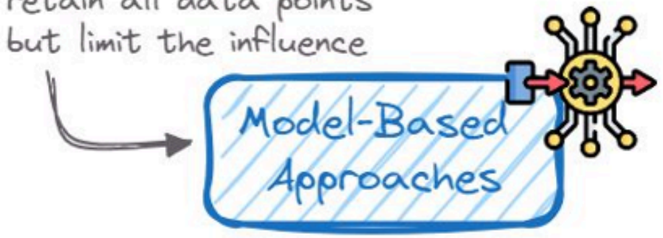
retain all data points
but limit the influence



assess the effect
on analysis results



retain all data points
but limit the influence



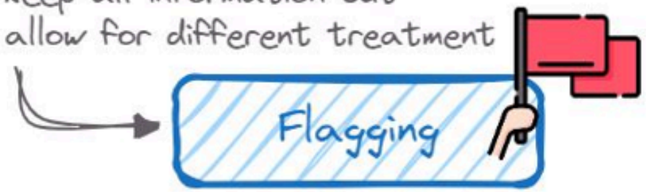
outliers are meaningful



analyze each subset separately



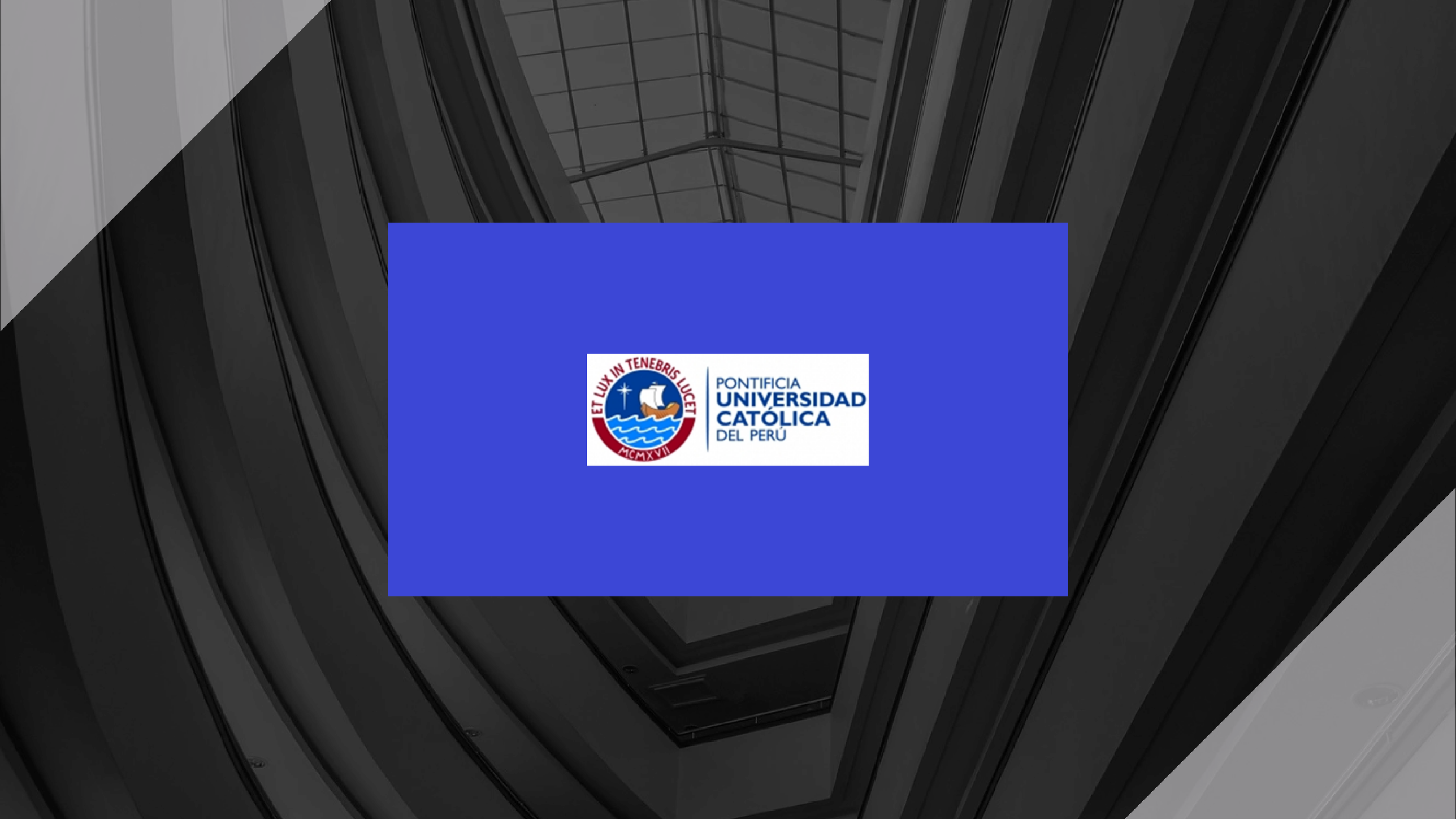
keep all information but
allow for different treatment



https://www.kaggle.com/discussions/general/465639

OUT PUT

"REPORTE DE PRE-PROCESAMIENTO"


ET LUX IN TENEBRIS LUCET
MCMXVII
PONTIFICIA
UNIVERSIDAD
CATÓLICA
DEL PERÚ